

PART IIB REVISION NOTES

4M24: Computational Statistics and Machine Learning

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1 Monte Carlo Methods

Central limit theorem for Monte Carlo: The Monte Carlo estimate follows

$$\frac{1}{N} \sum_n f(x_n) \sim \mathcal{N}\left(\mathbb{E}[f(x)], \frac{\sigma_f^2}{N}\right) \text{ as } N \rightarrow \infty$$

where σ_f^2 is the variance of $f(x)$.

**Proof:* Define a transform to zero mean and variance

$$\tilde{f}(x) = \frac{f(x) - \mathbb{E}[f(x)]}{\sigma_f}$$

Define $S_N = \frac{1}{N} \sum_{n=1}^N \tilde{f}(x_n)$ and $Z_N = \frac{S_N}{\sqrt{N}}$. The moment generating function of Z_N is

$$\begin{aligned} M_{Z_N}(t) &= \mathbb{E}[\exp(tZ_N)] = \mathbb{E}\left[\exp\left(t \frac{S_N}{\sqrt{N}}\right)\right] \\ &= \mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \sum_{n=1}^N \tilde{f}(x_n)\right)\right] \\ &= \prod_{n=1}^N \mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x_n)\right)\right] = \left(\mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x)\right)\right]\right)^N \end{aligned}$$

Taylor expansion of the exponential function gives

$$\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x)\right) = 1 + \frac{t}{\sqrt{N}} \tilde{f}(x) + \frac{t^2}{2N} \tilde{f}(x)^2 + O(N^{-2})$$

Since $\mathbb{E}[\tilde{f}(x)] = 0$ and $\mathbb{E}[\tilde{f}(x)^2] = 1$, the expectation is

$$\mathbb{E}\left[\exp\left(\frac{t}{\sqrt{N}} \tilde{f}(x)\right)\right] = 1 + \frac{t^2}{2N} + O(N^{-2})$$

Hence, the MGF of Z_N is

$$M_{Z_N}(t) = \left(1 + \frac{t^2}{2N} + O(N^{-2})\right)^N \rightarrow \exp\left(\frac{t^2}{2}\right) \text{ as } N \rightarrow \infty$$

which is the MGF of the standard normal. Hence $Z_N \rightarrow \mathcal{N}(0, 1)$, $S_N \rightarrow \mathcal{N}(0, \frac{1}{N})$, $\frac{1}{N} \sum_{n=1}^N f(x_n) \rightarrow \mathcal{N}(\mathbb{E}[f(x)], \frac{\sigma_f^2}{N})$.

Importance sampling: use another probability density $q(x)$ that satisfy $q(x) \neq 0$ when $f(x) \neq 0$ to sample x . The expectation can be rewritten as

$$\begin{aligned}\mathbb{E}_p[f(x)] &= \int f(x)p(x)dx \\ &= \int f(x)\frac{p(x)}{q(x)}q(x)dx \\ &= \mathbb{E}_q\left[f(x)\frac{p(x)}{q(x)}\right]\end{aligned}$$

The original Monte Carlo estimate $\frac{1}{N} \sum_n f(x_n)$ with each $x_n \sim p(x)$ becomes $\frac{1}{N} \sum_n f(x_n) \frac{p(x_n)}{q(x_n)}$ with each $x_n \sim q(x)$. The difference in variances is

$$\sigma_f^2 - \sigma_{\hat{f}}^2 = \int f(x)^2 p(x)dx - \int f(x)^2 \frac{p(x)^2}{q(x)}dx$$

The biggest decrease in variance is achieved when $q(x) \propto |f(x)|p(x)$, leading to $\sigma_{\hat{f}}^2 = 0$.

Control variates: use another function $g(x)$ with known expectation $\mathbb{E}[g(x)]$ to reduce the variance of the Monte Carlo estimate. The new estimator is

$$\hat{E}_f^c = \hat{E}_f + c(\hat{E}_g - \mathbb{E}[g(x)])$$

where $\hat{E}_f$ and $\hat{E}_g$ are the Monte Carlo estimates of $f(x)$ and $g(x)$, respectively.

The optimal value of c is

$$c^* = -\frac{\text{Cov}(\hat{E}_f, \hat{E}_g)}{\text{Var}(\hat{E}_g)}$$

giving the optimal estimator variance of

$$\text{Var}(\hat{E}_f^{c^*}) = \text{Var}(\hat{E}_f) - \frac{\text{Cov}(\hat{E}_f, \hat{E}_g)^2}{\text{Var}(\hat{E}_g)}$$

For multiple control variates, the estimator is $\hat{E}_f^c = \hat{E}_f + \sum_i c_i(\hat{E}_{g_i} - \mathbb{E}[g_i(x)])$, and the optimal coefficients are $\mathbf{c}^* = -C^{-1}\mathbf{b}$, where C is the covariance matrix of $\hat{E}_{g_i}$ and $\mathbf{b}$ is the covariance vector between $\hat{E}_f$ and $\hat{E}_{g_i}$. The optimal estimator variance is $\text{Var}(\hat{E}_f^{c^*}) = \text{Var}(\hat{E}_f) - \mathbf{b}^\top C^{-1}\mathbf{b}$.

2 Lebesgue Integral and Measure

Riemann integral: Define two summations for a function f on an interval $[a, b]$ and $\Delta x_k = [x_k, x_{k+1}]$:

$$\begin{aligned}S_p(x) &= \sum_{k=1}^n M_k(x_{k+1} - x_k) \text{ with } M_k = \sup_{x \in \Delta x_k} f(x) \\ s_p(x) &= \sum_{k=1}^n m_k(x_{k+1} - x_k) \text{ with } m_k = \inf_{x \in \Delta x_k} f(x)\end{aligned}$$

Define the limit of the summations as $n \rightarrow \infty$: $S(f) = \liminf_{n \rightarrow \infty} S_p(x)$, $s(f) = \limsup_{n \rightarrow \infty} s_p(x)$. If $S(f) = s(f) = A$, then A is the Riemann integral $\int_a^b f(x)dx$.

Limitations:

- The function must be continuous almost everywhere.
- The construction of the integral requires a partition of the domain.
- The exchange of limit and integral is not always valid.

Convergence of functions:

- *Pointwise convergence*: Given $\epsilon > 0$, there exists a natural number $N(\epsilon, x)$ such that for $n > N$, $|f_n(x) - f(x)| < \epsilon$. Does not preserve continuity.
- *Uniform convergence*: Given $\epsilon > 0$, there exists a natural number $N(\epsilon)$ such that for $n > N$, $|f_n - f| < \epsilon, \forall x \in D$. Preserves continuity and *allows exchange of limit and integral*.

Lebesgue integral: Partition the range of the function instead of the domain. For $0 \leq f_{\min} \leq f(x) \leq f_{\max}$, the range is partitioned into $f_{\min} = f_1, f_2, \dots, f_n = f_{\max}$. For each f_k , there are sets of values X_k such that $f_k \leq f(x) < f_{k+1}$ for $x \in X_k$, and the size (measure) of each set is $\mu(X_k)$.

The Lebesgue integral is the limiting value of the sum:

$$\int f d\mu = \lim_{n \rightarrow \infty} \sum_{k=1}^n f_k \mu(X_k) = \lim_{n \rightarrow \infty} \sum_{\max |f_k - f_{k-1}| \rightarrow 0}^n f_k \mu(X_k)$$

Advantages:

- Only requires a measure of sets on the domain of integration – more general
- Change of order of limit and integration only requires pointwise convergence.
- A Riemann integrable function is also Lebesgue integrable, but not vice versa.

Measure of sets: A set X and a set Σ generated from X (contains $X, \emptyset$, and other subsets of X , closed under countable complementation, union and intersections) defines a *measurable set* $\{X, \Sigma\}$. Along with a measure μ it defines a *measure space* $\{X, \Sigma, \mu\}$.

Rules for measure:

- If $\mu(E) \geq 0$ for all $E \in \Sigma$, all elements in Σ are measurable.
- $\mu(\emptyset) = 0$.
- If $E_i \subseteq E_j$, then $\mu(E_i) \leq \mu(E_j)$. (Monotonicity)
- $\mu(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mu(E_i)$ for disjoint sets $E_i \in \Sigma$. (Countable additivity)
- If (X, Σ_X) and (Y, Σ_Y) are both measurable, then a function $f : X \rightarrow Y$ is measurable if $f^{-1}(B) \in \Sigma_X$ for all $B \in \Sigma_Y$.

Lebesgue measure:

- For a continuous line segment on $\mathbb{R}$, the Lebesgue measure is the length of the segment. e.g. $I = [a, b]$, $\mu(I) = b - a$.

- The Lebesgue measure of a single point is zero. By additivity, the Lebesgue measure for the set of rational numbers $\mu(\mathbb{Q}) = 0$.
- The Lebesgue measure of countable sets (any set that can be put into a one-to-one correspondence with the set of integers $\mathbb{Z}$) is zero.

Probability measures: The *probability space* $\{\Omega, \mathcal{F}, P\}$ consists of three elements:

- Sample space Ω : the set of all possible outcomes.
- σ -algebra $\mathcal{F}$: certain subsets of Ω with outcomes to be considered.
- Probability measure P : a function that assigns a probability to each event in $\mathcal{F}$, satisfying $P(\Omega) = 1$ and countable additivity.

Random variables: For a probability space $\{\Omega, \mathcal{F}, P\}$ and a measurable space $\{E, \mathcal{E}\}$, there is an associated measurable function $X : \Omega \rightarrow E$ that satisfies $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{E}$. If $E = \mathbb{R}$, then $X : \Omega \rightarrow \mathbb{R}$ is a real-valued *random variable* such that $\{\omega \in \Omega : X(\omega) \leq r\} \in \mathcal{F}$ for every $r \in \mathbb{R}$.

The expectation of a random variable X is defined as $\mathbb{E}[X] = \int_{\Omega} X(\omega) dP(\omega)$.

Radon-Nikodym theorem: Consider two measures μ and ν on a measurable space $\{X, \Sigma\}$. If for every $A \in \Sigma$ where $\nu(A) = 0$ also have $\mu(A) = 0$, then μ is *absolutely continuous* with respect to ν , denoted as $\mu \ll \nu$.

Given a probability measure P , a new probability measure Q can be defined if there exists a function $L(\omega)$ such that $dQ(\omega) = L(\omega)dP(\omega)$.

For any measurable set B , $Q(B) = \int_B L(\omega)dP(\omega)$ is the probability of B under the new measure Q .

The expectation of a function $F(\omega)$ can be computed as

$$\int_{\Omega} F(\omega) dQ(\omega) = \int_{\Omega} F(\omega) L(\omega) dP(\omega)$$

where the function $L(\omega) = \frac{dQ(\omega)}{dP(\omega)}$ is the Radon-Nikodym derivative if $Q \ll P$, subject to (1) $L(\omega) \geq 0$ for all $\omega \in \Omega$ and (2) $\int_{\Omega} L(\omega) dP(\omega) = 1$.

For a random variable X on $\mathbb{R}$, the probability of a small neighborhood B around the outcome $X = x \in \mathbb{R}$ is $\int_B dP = \int_B L(x) d\mu(x) = \int_B L(x) dx$. The function $L(x) = \frac{dP(x)}{dx}$ is the probability density function of P with respect to the Lebesgue measure.

3 Markov Chain Monte Carlo

Probability inversion: For data $\mathbf{x} \in \mathcal{X}$ and parameters $\boldsymbol{\theta} \in \Theta$, the posterior distribution is $\pi(\boldsymbol{\theta}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})}{\int_{\Theta} p(\mathbf{x}|\boldsymbol{\theta})\pi_0(\boldsymbol{\theta})d\boldsymbol{\theta}}$. Inference requires integration with respect to the posterior

$$I(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}|\mathbf{x}}[f(\boldsymbol{\theta})] = \int_{\Theta} f(\boldsymbol{\theta})\pi(\boldsymbol{\theta}|\mathbf{x})d\boldsymbol{\theta}$$

The Monte-Carlo estimate is

$$\hat{I}_N(\boldsymbol{\theta}) = \frac{1}{N} \sum_{n=1}^N f(\boldsymbol{\theta}_n) \text{ with } \boldsymbol{\theta}_n \sim \pi(\boldsymbol{\theta}|\mathbf{x})$$

The estimator is unbiased and consistent as $N \rightarrow \infty$, $\hat{I}_N(\boldsymbol{\theta}) \rightarrow I(\boldsymbol{\theta})$ and the error $\sqrt{N}(\hat{I}_N(\boldsymbol{\theta}) - I(\boldsymbol{\theta})) \rightarrow \mathcal{N}(0, \sigma_f^2)$.

Markov chains: stochastic transitions governed by the Markov property

$$p(\boldsymbol{\theta}^{(i)} | \boldsymbol{\theta}^{(i-1)}, \dots, \boldsymbol{\theta}^{(1)}) = T(\boldsymbol{\theta}^{(i)} | \boldsymbol{\theta}^{(i-1)})$$

Invariant distribution: the Markov chain converges to the invariant distribution $\pi(\boldsymbol{\theta}^{(\infty)}) \rightarrow \pi(\boldsymbol{\theta})$ irrespective of the initial state.

Irreducible: the Markov chain can reach any state from any other state in a finite number of steps.

Aperiodic: the Markov chain does not get trapped in cycles.

Detailed balance (reversible): a Markov chain satisfies detailed balance if $\pi(\boldsymbol{\theta}^{(i)})T(\boldsymbol{\theta}^{(i-1)} | \boldsymbol{\theta}^{(i)}) = \pi(\boldsymbol{\theta}^{(i-1)})T(\boldsymbol{\theta}^{(i)} | \boldsymbol{\theta}^{(i-1)})$. This is a sufficient condition for an invariant distribution to exist.

3.1 Metropolis-Hastings Algorithm

Define the *transition kernel* as

$$\underbrace{P(x, dy)}_{\text{prob. of transition from } x \text{ to } dy} = \underbrace{p(x, y)dy}_{\text{prob. of moving to } dy \text{ around } y} + \underbrace{r(x)\delta_x(dy)}_{\text{prob. of staying at } x}$$

If $p(x, x) = 0$ and $\int_{\mathbb{R}^d} P(x, dy) = 1$, then $r(x) = 1 - \int_{\mathbb{R}^d} p(x, y)dy$ is the probability that the chain remains at x .

If $p(x, y)$ satisfies detailed balance $\pi(x)p(x, y) = \pi(y)p(y, x)$, then $\pi(\cdot)$ is the invariant density of the transition kernel $P(x, \cdot)$. In this case the transition kernel is reversible, i.e. $P(x, dy)\pi(dx) = P(y, dx)\pi(dy)$.

**Proof:*

$$\begin{aligned} \int_{\mathbb{R}^d} P(x, A)\pi(x)dx &= \int_{\mathbb{R}^d} \left[\int_A p(x, y)dy \right] \pi(x)dx + \int_{\mathbb{R}^d} r(x)\delta_x(A)\pi(x)dx \\ &= \int_A \left[\int_{\mathbb{R}^d} \pi(x)p(x, y)dx \right] dy + \int_A r(x)\pi(x)dx \\ &= \int_A \left[\int_{\mathbb{R}^d} \pi(y)p(y, x)dx \right] dy + \int_A r(x)\pi(x)dx \\ &= \int_A [1 - r(y)]\pi(y)dy + \int_A r(y)\pi(y)dy \\ &= \int_A \pi(y)dy = \pi^*(A) \end{aligned}$$

To achieve this, the *proposal density* $q(x, y)$ is balanced with an *acceptance probability* that satisfies $\pi(x)q(x, y)\alpha(x, y) = \pi(y)q(y, x)\alpha(y, x)$. Set $\alpha(y, x) = 1$ then $\alpha(x, y) = \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}$ if $\pi(x)q(x, y) \geq \pi(y)q(y, x)$; otherwise $\alpha(x, y) = 1$. This gives $p_{\text{MH}}(x, y) = q(x, y)\alpha(x, y)$ for $x \neq y$ with the acceptance probability

$$\alpha(x, y) = \min \left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} \right)$$

The transition kernel is

$$P_{\text{MH}}(x, dy) = q(x, y)\alpha(x, y)dy + \left(1 - \int_{\mathbb{R}^d} q(x, y)\alpha(x, y)dy\right) \delta_x(dy)$$

Algorithm 1 Metropolis-Hastings Algorithm

```

Initialize  $x^{(1)}$ 
for  $j = 1 \rightarrow N$  do
    Simulate  $y \sim q(x^{(j)}, \cdot)$            ▷ Propose a candidate from proposal distribution
    Simulate  $u \sim U(0, 1)$            ▷ Generate a random number for acceptance decision
    if  $u \leq \alpha(x^{(j)}, y)$  then       ▷ Acceptance probability  $\alpha(x, y) = \min\left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right)$ 
        Set  $x^{(j+1)} = y$                ▷ Accept the proposal and move to the new state
    else
        Set  $x^{(j+1)} = x^{(j)}$            ▷ Reject the proposal and stay at the current state
    end if
end for
Return  $\{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$ 

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3.2 Gibbs Sampling

Product transition kernels: the product kernel $P_1(x_1, dy_1|x_2)P_2(x_2, dy_2|y_1)$ has invariant density $\pi(y_1, y_2)$ if:

- the conditional kernel $P_1(x_1, dy_1|x_2)$ has invariant density $\pi_{1|2}(\cdot|x_2)$ for fixed x_2
- the conditional kernel $P_2(x_2, dy_2|x_1)$ has invariant density $\pi_{2|1}(\cdot|x_1)$ for fixed x_1 .

**Proof:*

$$\begin{aligned}
& \int \int P_1(x_1, dy_1|x_2)P_2(x_2, dy_2|y_1)\pi(x_1, x_2)dx_1dx_2 \\
&= \int P_2(x_2, dy_2|y_1) \left[\int P_1(x_1, dy_1|x_2)\pi_{1|2}(x_1|x_2)dx_1 \right] \pi_2(x_2)dx_2 \\
&= \int P_2(x_2, dy_2|y_1)\pi_{1|2}^*(dy_1|x_2)\pi_2(x_2)dx_2 \\
&= \int P_2(x_2, dy_2|y_1)\pi_{2|1}(x_2|y_1)\pi_1^*(dy_1)dx_2 \\
&= \pi_1^*(dy_1) \int P_2(x_2, dy_2|y_1)\pi_{2|1}(x_2|y_1)dx_2 \\
&= \pi_1^*(dy_1)\pi_{2|1}^*(dy_2|y_1) = \pi^*(dy_1, dy_2)
\end{aligned}$$

Gibbs sampler: set kernels to $P_1(x_1, dy_1|x_2) = \pi_{1|2}^*(dy_1|x_2)$ and $P_2(x_2, dy_2|x_1) = \pi_{2|1}^*(dy_2|x_1)$. Use exact conditional densities as the proposal densities, i.e. for transition of x_1 with x_2 fixed, $q([x_1, x_2], [y_1, y_2]) = \pi(y_1|x_2)$ and $q([y_1, y_2], [x_1, x_2]) = \pi(x_1|y_2)$. As a result, the acceptance probability is $\alpha = 1$ for all moves.

Limitation: small step sizes and poor mixing if the components are highly correlated.

Algorithm 2 Gibbs Sampler

```
1: for  $i = 1, \dots, N$  do
2:   for  $j = 1, \dots, d$  do ▷ Update one dimension at a time
3:      $x_j^{(i)} \sim \pi(\cdot \mid x_1^{(i)}, \dots, x_{j-1}^{(i)}, x_{j+1}^{(i-1)}, \dots, x_d^{(i-1)})$  ▷ Sample from the conditionals
4:   end for
5: end for
6: return  $x^{(0)}, \dots, x^{(N)}$ 
```

Mixture transition kernels: mixing transitional kernels that have the same invariant density, i.e. $P(x, dy) = \sum_i w_i P_i(x, dy)$ with $\sum_i w_i = 1$.

Advantage: for multimodal distributions, the mixture kernel can have better mixing properties by combining local and global moves.

3.3 Data Augmentation

Augment a target density $\pi(\theta)$ with ϕ to form $\pi(\theta, \phi)$. Draw samples $\theta^{(n)}, \phi^{(n)} \sim \pi(\theta, \phi)$, then for each $\theta^{(n)}$ it follows $\int \pi(\theta^{(n)}, \phi) d\phi = \pi(\theta^{(n)})$.

4 Vector, Metric and Banach Spaces

Vector space V : a collection of vectors x with axioms (group):

- $x + y = y + x$ (commutativity)
- $(x + y) + z = x + (y + z)$ (associativity)
- There exists a zero vector such that $x + 0 = x$ for all x
- For each x , there exists an additive inverse such that $x + (-x) = 0$
- For $\alpha \in \mathbb{F}$, $\alpha x \in V$ for all $x \in V$ (scalar multiplication is closed)
- $\alpha(x + y) = \alpha x + \alpha y$ (distributivity)
- $\alpha(\beta x) = (\alpha\beta)x$ (associativity of scalar multiplication)
- $0 \times x = 0, 1 \times x = x$

Inner product space: space with an inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ satisfying:

- $\langle x, y \rangle = \overline{\langle y, x \rangle}$
- $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$ for $\alpha, \beta \in \mathbb{R}$
- $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$
- For $a, b \in \mathbb{C}^n$, $\langle a, b \rangle = \sum_{i=1}^n a_i^* b_i$

Inner product of functions: $\langle f, g \rangle = \int_a^b f(x)g(x)dx$ for $x \in [a, b] \in \mathbb{R}$.

Vector norm: the length of a vector, satisfying:

- Schwarz inequality: $|\langle x, y \rangle| \leq \|x\| \|y\|$

- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$
- Parallelogram law: $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$

Orthogonality: two elements x and y are orthogonal if and only if $\langle x, y \rangle = 0$. e.g. $\sin(x)$ and $\cos(x)$ are orthogonal on $[-\pi, \pi]$.

Orthonormal basis: For a vector space V , a *basis* is a *finite* set of linearly independent, orthogonal unit vectors that can express any vector in V as $x = \sum_{i=1}^n \alpha_i e_i$.

Cauchy sequence: a sequence of vectors $\{x_n\}$ is a Cauchy sequence if for any $\epsilon > 0$, there exists a natural number N such that for all $m, n > N$, $\|x_n - x_m\| < \epsilon$.

If every Cauchy sequence in a vector space converges to a limit that is *also in the vector space*, then the vector space is *complete*.

Metric vector space: a vector space with a distance function $\rho(x, y)$ that maps two vectors to a non-negative real number, satisfying:

- $\rho(x, y) = 0$ if and only if $x = y$
- $\rho(x, y) = \rho(y, x)$ (symmetry)
- $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$ (triangle inequality)

Normed space: a metric space with distance function $\rho(x, y) = \|x - y\|$ induced by a non-negative norm $\|\cdot\|$ for all $x, y \in V$.

Banach space: a *complete normed space*. All finite-dimensional normed spaces are Banach spaces.

The l_p space is a Banach space where $\sum_{i=1}^{\infty} |x_i|^p < \infty$ for $p \geq 1$. The norm is the l^p norm $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$.

The L_p space is a Banach space where $\int_a^b |f(x)|^p dx < \infty$, where dx is the Lebesgue measure. The norm is the L^p norm $\|f\|_p = \left(\int_a^b |f(x)|^p dx\right)^{1/p}$. The L^p norm can be used to measure the complexity of a machine learning model.

5 Hilbert Spaces

Pre-Hilbert space: an incomplete inner product space.

Hilbert space: a *complete inner product space*.

Any f in a Hilbert space can be expressed as $f = \sum_{i=1}^{\infty} \langle f, e_i \rangle e_i$. The infinite set of infinite-dimensional orthonormal vectors $\{e_i\}$ can be a basis if and only if

$$\{e_i\} \text{ is complete} \iff \|f\|^2 = \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 \quad (\text{Parseval's identity})$$

$*l^2$ and L^2 spaces are both Hilbert spaces.

The norm of a Hilbert space is induced by the inner product $\|f\| = \sqrt{\langle f, f \rangle}$. The L^2 norm $\|f - g\|_2$ measures the closeness of functions in the mean but not necessarily

pointwise. When $\|f - g\|_2 = 0$, there is *equivalence almost everywhere* except for sets of measure zero. Hence, the L^2 space is a space of *equivalence classes of functions*.

Equivalence of l^2 and L^2 spaces: a function $f \in L^2$ can be represented with a sequence $\{c_k = \langle f, \phi_k \rangle\} \in l^2$ by Fourier series

$$f = \sum_{k=1}^{\infty} c_k \phi_k$$

where $\{\phi_k\}$ is an orthonormal basis of L^2 .

l^2 and L^2 are isomorphic, i.e. there is a one-to-one correspondence between the elements of l^2 and L^2 . l^2 is a coordinate system for L^2 .

Reproducing kernel Hilbert space (RKHS): a Hilbert space of functions where point evaluation is continuous i.e. functions close in norm are also close pointwise.

The *reproducing kernel function* $K_x = K(x, \cdot)$ reproduces function evaluation by inner product $f(x) = \langle f(\cdot), K_x(\cdot) \rangle$. The kernel function is symmetric and positive definite, and uniquely defines the RKHS.

Function approximation in RKHS: Given data $\{(x_i, y_i)\}_{i=1}^n$ where $y_i = f(x_i)$, the regularised empirical error

$$\frac{1}{N} \sum_{n=1}^N |y_n - f(x_n)|^2 + \lambda \|f\|^2$$

is minimised by approximating functions of the form

$$\hat{f}(\cdot) = \sum_{n=1}^N \alpha_n K(x_n, \cdot)$$

**Solution:*

$$\begin{aligned} \sum_{n=1}^N |y_n - \hat{f}(x_n)|^2 &= \sum_{n=1}^N \left| y_n - \sum_{m=1}^N \alpha_m K(x_m, x_n) \right|^2 = (K\alpha - y)^\top (K\alpha - y) \\ \|\hat{f}\|^2 &= \langle \hat{f}, \hat{f} \rangle = \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m \langle K(x_n, \cdot), K(x_m, \cdot) \rangle = \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m K(x_n, x_m) = \alpha^\top K \alpha \end{aligned}$$

The third equality follows from the reproducing property of the kernel $f = \langle f, k \rangle$.

Taking derivative with respect to α gives:

$$\frac{\partial}{\partial \alpha} \left[\frac{1}{N} (K\alpha - y)^\top (K\alpha - y) + \lambda \alpha^\top K \alpha \right] = \frac{2}{N} K(K\alpha - y) + 2\lambda K \alpha$$

Setting the derivative to zero

$$\begin{aligned} \frac{1}{N} K(K\alpha - y) + \lambda K \alpha &= 0 \\ (K + N\lambda I) \alpha &= y \\ \alpha &= (K + N\lambda I)^{-1} y \end{aligned}$$

The term $\lambda \|f\|^2$ penalises the complexity of the function and prevents overfitting.

6 Sobolev Spaces

Sobolev space: a Sobolev space $W_p^k(\Omega)$ is a Banach space where functions $f \in L^p(\Omega)$ have *weak derivatives* $D^s f \in L^p(\Omega)$ for $0 \leq s \leq k$. The norm is defined as

$$\|f\|_{W_p^k} = \left(\sum_{s=0}^k \int_{\Omega} |D^s f(x)|^p dx \right)^{1/p}$$

For $p = 2$, the Sobolev space $W_2^k(\Omega)$ is a Hilbert space with inner product

$$\langle f, g \rangle_{W_2^k} = \sum_{s=0}^k \langle D^s f, D^s g \rangle = \sum_{s=0}^k \int_{\Omega} D^s f(x) D^s g(x) dx$$

The norm and the inner product in $W_2^k(\Omega)$ encodes the smoothness of function.

Weak derivatives: for functions $f : [0, 1] \rightarrow \mathbb{R}$, a function $h : [0, 1] \rightarrow \mathbb{R}$ is a weak derivative of f if for every differentiable function $g : [0, 1] \rightarrow \mathbb{R}$ with $g(0) = g(1) = 0$,

$$\int_0^1 f(x) g'(x) dx = - \int_0^1 h(x) g(x) dx$$

Function approximation by polynomial expansion: By Fourier expansion, functions from the set $C_2[-\pi, \pi] = C[-\pi, \pi] \cap L^2[-\pi, \pi]$ can be represented as $f(x) = \sum_{k=0}^{\infty} c_k \exp(ikx)$ with $c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \exp(-ikx) dx$. The L^2 norm of f is $\|f\|^2 = \sum_{k=0}^{\infty} c_k^2$.

The approximating functions $f_n(x) = \sum_{k=0}^n c_k \exp(ikx)$ are in the Sobolev space W_2^s with the norm $\|f\|_{W_2^s}^2 = \|D^s f\|_{L^2}^2 = \sum_{k=0}^{\infty} k^{2s} c_k^2$.

*The approximation error is bounded as

$$\begin{aligned} \|f - f_n\|_{L^2}^2 &= \sum_{k=n+1}^{\infty} c_k^2 = \sum_{k=n+1}^{\infty} \frac{1}{k^{2s}} k^{2s} c_k^2 \\ &< \frac{1}{n^{2s}} \sum_{k=n+1}^{\infty} k^{2s} c_k^2 < \frac{1}{n^{2s}} \sum_{k=1}^{\infty} k^{2s} c_k^2 \\ &= \frac{1}{n^{2s}} \|f\|_{W_2^s}^2 \end{aligned}$$

As s increases, the approximation error decreases faster with increasing n .

For d -dimensional domains, the error bound is

$$\|f - f_n\|_{L^2}^2 < \frac{1}{dn^{\frac{2s}{d}}} \|f\|_{W_2^s}^2$$

which shows the curse of dimensionality – convergence rate decreases as d increases.

7 Gaussian Measures in Hilbert Space

Lebesgue measure in D dimensions: $\mu = \prod_{d=1}^D \mu_d$.

There is no Lebesgue measure in infinite dimensional Hilbert or Banach spaces.

*Proof:

- A Hilbert space H has an associated basis $\{e_i\}_{i=1}^{\infty}$.
- Denote a ball of radius r with center at x as $B(x, r)$.
- For a ball $B(0, 2)$, place a ball of radius $\frac{1}{2}$ at each basis vector e_i : $\{B(e_i, \frac{1}{2})\}_{i=1}^{\infty}$.
- The balls do not intersect with each other: $B(e_i, \frac{1}{2}) \cap B(e_j, \frac{1}{2}) = \emptyset$ for $i \neq j$.
- And the union of all balls is a subset of $B(0, 2)$: $\bigcup_{i=1}^{\infty} B(e_i, \frac{1}{2}) \subset B(0, 2)$.
- Since the Lebesgue measure in infinite dimensions is *translation invariant*, $\mu(B(e_i, \frac{1}{2})) = \mu(B(e_j, \frac{1}{2}))$ for all i, j .
- *Monotonicity and countable additivity* gives $\sum_{i=1}^{\infty} \mu(B(e_i, \frac{1}{2})) \leq \mu(B(0, 2))$.
- The Lebesgue measure $\mu(B(0, 2)) < \infty$, but $\sum_{i=1}^{\infty} \mu(B(e_i, \frac{1}{2})) = \infty$, which is a contradiction. Therefore the Lebesgue measure is not well-defined in infinite dimensional Hilbert spaces.

Gaussian measure: The standard Gaussian measure is

$$g(B) = \frac{1}{\sqrt{2\pi}} \int_B \exp\left(-\frac{x^2}{2}\right) dx$$

For a product measure $\mu = \prod_{d=1}^{\infty} g_k$ on $\mathbb{R}^{\infty}$ to be well defined, $\exp\left(-\frac{1}{2} \sum_{k=1}^{\infty} x_k^2\right)$ is finite and non-zero. So $\sum_{k=1}^{\infty} x_k^2 < \infty$, which is the definition of l_2 :

$$x \in l_2 = \{x \in \mathbb{R}^{\infty} : \sum_{k=1}^{\infty} x_k^2 < \infty\}$$

Gaussian measure in infinite dimensions: A Gaussian measure is a probability measure characterised by a mean function and a covariance operator.

Diagonal covariance operator: $S_x = (a_n x_n)_{n=1}^{\infty}$ for $x \in l^2$ subject to $\sum_{k=1}^{\infty} a_k < \infty$.

Full covariance operators: need to be *trace class*, i.e. the trace $\text{Tr}(C)$ is finite

$$\text{Tr}(C) = \sum_{n=1}^{\infty} \langle e_n, C e_n \rangle < \infty$$

The inner product $\langle e_n, C e_n \rangle$ is

$$\begin{aligned} \langle e_n, C e_n \rangle &= \mathbb{E}[(e_n, u)(u, e_n)] \\ &= \mathbb{E} \left[\int_{\Omega} \int_{\Omega} e_n(x) [u(x)u(y)] e_n(y) dy dx \right] \\ &= \mathbb{E} \left[\int_{\Omega} e_n(x) \left(\int_{\Omega} u(x)u(y) e_n(y) dy \right) dx \right] \\ &= \int_{\Omega} e_n(x) \left(\int_{\Omega} \mathbb{E}[u(x)u(y)] e_n(y) dy \right) dx \\ &= \int_{\Omega} e_n(x) \left(\int_{\Omega} c(x, y) e_n(y) dy \right) dx \end{aligned}$$

where $c(x, y) = \mathbb{E}[u(x)u(y)]$ is the covariance function and $C e_n(x) = \int_{\Omega} c(x, y) e_n(y) dy$.

Gaussian measures in Hilbert space: In a Hilbert space, two Gaussian measures are either equivalent or singular.

Equivalent (mutually absolutely continuous): Two measures are equivalent if they both assign zero measure to the same sets.

Singular: Two measures are singular w.r.t each other if one measure assigns zero to a set and the other measure assigns zero measure to the set complement.

For two Gaussian measures $\mathcal{N}(0, C)$ and $\mathcal{N}(v, C)$ in l^2 , two measures are equivalent if $v \in \text{Image}(C^{1/2})$ i.e. $v = C^{1/2}u$ for some $u \in l^2$, and singular otherwise. The subspace for v is the Cameron-Martin space (aka the reproducing kernel Hilbert space).

Bayes' rule in Hilbert space: The Radon-Nikodym derivative of the posterior probability measure with respect to the prior probability measure is given by

$$\frac{d\mu^y}{d\mu^0} \propto \underbrace{P(y|x)}_{\text{likelihood}}$$

In a infinite dimensional Hilbert space, the prior measure over functions is a Gaussian measure $\mu_0 = \mathcal{N}(m, C)$ – formal definition of Gaussian Process prior.

8 MCMC in Infinite-Dimensional Spaces

Metropolis-Hastings algorithm in infinite dimensions: Define a proposal

$$v = \sqrt{1 - \beta^2}u + \beta\xi, \xi \sim \mathcal{N}(0, C)$$

such that u and v both follow $\mathcal{N}(0, C)$. Given a prior Gaussian measure $\mu^0 = \mathcal{N}(0, C)$ and a likelihood $P(y|v) = \exp(-\Phi(v))$, the acceptance probability (replacing $\pi(v)q(v, u)$ with $\mu^y(dv)q(dv, u)$, where μ^y is the posterior measure) is

$$\alpha(u, v) = \min \left(1, \frac{\mu^y(dv)q(v, du)}{\mu^y(du)q(u, dv)} \right) = \min \left(1, \frac{P(y|v)}{P(y|u)} \right) = \min(1, \exp(\Phi(u) - \Phi(v)))$$

For $v = u + \beta\xi, \xi \sim \mathcal{N}(0, C)$ the acceptance probability is $\min(1, \exp(J(v) - J(u)))$, where $J(v) = -\Phi(v) - \frac{1}{2}|C^{-1/2}v|^2$. The expectation of the norm $\mathbb{E}[|C^{-1/2}v|^2]$ is infinite.

9 Langevin Dynamics

Langevin stochastic differential equation (SDE):

$$dX_t = -\nabla_x U(X_t)dt + \sqrt{2}dB_t = \nabla_x \log \pi(X_t)dt + \sqrt{2}dB_t$$

where $U(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}$ is a potential function, $(B_t)_{t \geq 0}$ is the Brownian motion and $\pi(x) \propto \exp(-U(x))$ is the stationary distribution of the SDE.

The measure of X_t, ρ_t , satisfies the Fokker-Planck equation

$$\frac{\partial \rho_t}{\partial t} = \nabla \cdot (\rho_t \nabla \log \pi) + \Delta \rho_t$$

with asymptotic solution $\rho_\infty = \pi$.

Log-concave: A probability measure with density π is log-concave if

$$\pi(\lambda x + (1 - \lambda)y) \geq \pi(x)^\lambda \pi(y)^{1-\lambda} \text{ for } 0 < \lambda < 1$$

A probability measure with density π is m-strongly log-concave if

$$\pi(\lambda x + (1 - \lambda)y) \geq \pi(x)^\lambda \pi(y)^{1-\lambda} \exp\left(\frac{m}{2}\lambda(1 - \lambda)\|x - y\|^2\right) \text{ for } 0 < \lambda < 1$$

i.e. there are no flat regions and the maximum is unique. For one dimension

$$\nabla^2 \log \pi(x) \leq -m < 0 \quad \forall x$$

Convergence of Langevin dynamics: For an m-strongly log-concave π , the convergence rate to the stationary measure is exponentially fast

$$W_2(\rho_t, \pi) \leq e^{-mt} \left\{ \|x - x^*\| + \left(\frac{d}{m}\right)^{\frac{1}{2}} \right\}$$

for $\pi_0 = \delta_x$ and $x^* = \arg \max_x \pi(x)$.

Metropolis-adjusted Langevin algorithm (MALA): Given a state of the Markov chain X_k , propose a new state $\bar{X}_{k+1}$ by first-order Euler discretisation with step size γ :

$$\bar{X}_{k+1} = X_k + \gamma \nabla \log \pi(X_k) + \sqrt{2\gamma} W_k, W_k \sim \mathcal{N}(0, I)$$

To correct the discretisation bias, accept the proposed move with probability

$$\alpha(X_k, \bar{X}_{k+1}) = \min \left(1, \frac{\pi(\bar{X}_{k+1})q(X_k|\bar{X}_{k+1})}{\pi(X_k)q(\bar{X}_{k+1}|X_k)} \right)$$

where $q(\bar{X}_{k+1}|X_k) \propto \exp \left(-\frac{1}{4\gamma} \|\bar{X}_{k+1} - X_k - \gamma \nabla \log \pi(X_k)\|^2 \right)$

Limitation: the acceptance probability is exponentially low for high dimension, leading to poor mixing.

Unadjusted Langevin algorithm (ULA): Run the first-order Euler discretisation without the Metropolis-Hastings correction, i.e. $X_{k+1} = X_k + \gamma \nabla \log \pi(X_k) + \sqrt{2\gamma} W_k$. The stationary distribution is a biased distribution π_γ , with asymptotic bias of order $\mathcal{O}(\gamma^{\frac{1}{2}})$.

Langevin schemes for Bayesian inference: For a target distribution $\pi(x) \propto p(x) \prod_{i=1}^n p(y_i|x)$, where $p(x)$ is the prior and $p(y_i|x)$ is the likelihood for data vector y_i , samples can be generated from

$$X_{k+1} = X_k + \gamma \left(\nabla \log p(X_k) + \sum_{i=1}^n \nabla \log p(y_i|X_k) \right) + \sqrt{2\gamma} W_k$$

For MALA, the complexity is $\mathcal{O}(n)$ for both the gradient and the acceptance probability (can be avoided in ULA) computation.

Stochastic gradient Langevin dynamics (SGLD): An unbiased estimate of the gradient can be computed more efficiently by subsampling the data. At each iteration k , choose a random index set $\mathcal{I}_k \subset \{1, \dots, n\}$ and approximate the gradient

$$\nabla \log \widehat{\pi}(X_k) = \nabla \log p(X_k) + \frac{n}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \nabla \log p(y_i | X_k)$$

The complexity is $\mathcal{O}(|\mathcal{I}_k|)$, and convergence rate is similar to ULA.

10 Energy-Based and Score-Based Models

Generative models: approximate the data distribution p_{data} with a model distribution p_{θ} , by learning parameters θ from data $\mathcal{D} = \{x_i\}_{i=1}^N$ drawn from p_{data} .

Energy-based models: The probability of a state x is defined as

$$p(x) = \frac{1}{Z_{\beta}} \exp(-\beta E(x))$$

where $E(x)$ is the energy function, β is the inverse temperature and $Z_{\beta} = \int \exp(-\beta E(x)) dx$ is the partition function. EBMs parameterise the energy function, giving $p_{\theta}(x) \propto \exp(-E_{\theta}(x))$.

The EBMs are trained via maximum likelihood or contrastive divergence.

Maximum likelihood: maximise the log-likelihood $\mathcal{L}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\log p_{\theta}(x)]$ via gradient methods, with gradient:

$$\nabla_{\theta} \mathcal{L}(\theta) = -\mathbb{E}_{x \sim p_{\text{data}}} [-\nabla_{\theta} E_{\theta}(x)] + \underbrace{\mathbb{E}_{x \sim p_{\theta}} [-\nabla_{\theta} E_{\theta}(x)]}_{\nabla_{\theta} \log Z_{\theta}}$$

Limitation: the second term requires sampling from the model distribution p_{θ} , which is computationally expensive.

Contrastive divergence:

1. For each x_i , initialise a MCMC chain with target distribution p_{θ} at $x_i^{(0)}$.
2. Run the MCMC chain for k steps to obtain $x_i^{(k)}$.
3. Compute the gradient estimate

$$\hat{\mathcal{L}} = -\frac{1}{N} \sum_{i=1}^N \left(-\nabla_{\theta} E_{\theta}(x_i^{(0)}) + \nabla_{\theta} E_{\theta}(x_i^{(k)}) \right)$$

4. Update parameters $\theta \leftarrow \theta + \eta \hat{\mathcal{L}}$.

Limitations: bias in the gradient estimate and MCMC can be very slow.

Score-based models: Learn to match the score function $s_{\theta}(x) = \nabla_x \log p_{\theta}(x) = -\nabla_x E_{\theta}(x)$, bypassing the partition function. Trained by score matching, which minimises the Fisher divergence between p_{data} and p_{θ} :

$$J(\theta) = \frac{1}{2} \mathbb{E}_{x \sim p_{\text{data}}} \left[\left\| s_{\theta}(x) - \underbrace{\nabla_x \log p_{\text{data}}(x)}_{\text{intractable}} \right\|^2 \right]$$

which can be rewritten as

$$J_{\text{SM}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} \left[\underbrace{\text{Tr}(\nabla_x s_{\theta}(x))}_{\text{expensive for high-dim.}} + \frac{1}{2} \|s_{\theta}(x)\|^2 \right]$$

Sliced score matching: approximate the trace of the Jacobian by projecting the score function onto random directions $v \sim \mathcal{N}(0, I)$

$$J_{\text{SSM}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}, v \sim \mathcal{N}(0, I)} \left[v^{\top} \nabla_x s_{\theta}(x) v + \frac{1}{2} \|s_{\theta}(x)\|^2 \right]$$

In practice, one sample of v is used for each data point:

$$\hat{J}_{\text{SSM}}(\theta) = \frac{1}{N} \sum_{i=1}^N \left[v_i^{\top} \nabla_x s_{\theta}(x_i) v_i + \frac{1}{2} \|s_{\theta}(x_i)\|^2 \right]$$

Sampling from generative models: use Langevin dynamics

$$dx_t = s_{\theta}(x)dt + \sqrt{2}dB_t$$

which have stationary distribution p_{θ} . In practice, discretise the SDE with step size γ :

$$x^{(k+1)} = x^{(k)} + \gamma s_{\theta}(x^{(k)}) + \sqrt{2\gamma} z^{(k)}, \quad z^{(k)} \sim \mathcal{N}(0, I)$$

Parallel tempering (replica exchange): For $\pi(x) = \exp(-U(x))$, define the tempered (smoothed) distribution $\pi_{\beta}(x) = \exp(-\beta U(x))$ with inverse temperature β . Run MCMC chains on multiple tempered distributions $\{\pi_{\beta_i}\}$ in parallel, and exchange samples between i -th and j -th chains with acceptance probability:

$$\alpha = \min \left(1, \frac{\pi_{\beta_i}(x_t^j) \pi_{\beta_j}(x_t^i)}{\pi_{\beta_i}(x_t^i) \pi_{\beta_j}(x_t^j)} \right)$$